

MIANYANG, China

WMO: 561960

Lat: 31.424N

Lon: 104.749E

Elev: 1713

StdP: 13.81

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 33.0	(c) 35.0	(d) 15.1	(e) 12.6	(f) 45.9	(g) 20.7	(h) 16.5	(i) 45.9	(j) 13.1	(k) 45.3	(l) 10.7	(m) 43.3	(n) 2.2	(o) 320	(p) 0.320

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	93.4	75.6	91.2	75.1	89.0	74.2	79.0	88.3	78.0	86.7	77.0	85.1	5.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.6	148.0	84.0	75.6	142.9	82.9	74.7	138.4	82.0	44.3	88.3	43.1	87.0	42.0	85.4	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.9	(b) 9.8	(c) 8.3	(e) 29.7	(f) 96.7	(g) 2.2	(h) 1.6	(i) 28.1	(j) 97.8	(k) 26.8	(l) 98.7	(m) 25.6	(n) 99.6	(o) 23.9	(p) 100.7		
(4) 11.9	(5) 9.8	(6) 8.3	(7) 29.7	(8) 96.7	(9) 2.2	(10) 1.6	(11) 28.1	(12) 97.8	(13) 26.8	(14) 98.7	(15) 25.6	(16) 99.6	(17) 23.9	(18) 100.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.6	43.4	47.9	56.1	65.6	72.5	76.9	80.6	80.2	72.7	64.6	55.5	45.9
	DBStd	13.83	4.14	5.34	6.45	6.35	4.78	4.27	3.99	4.62	4.98	4.72	5.11	4.09
	HDD50	468	209	94	16	1	0	0	0	0	0	0	13	135
	HDD65	2447	671	478	284	68	4	0	0	0	2	64	285	591
	CDD50	5424	3	36	205	469	697	807	950	936	680	454	178	9
	CDD65	1928	0	0	8	86	236	357	485	471	232	53	0	0
	CDH74	15654	0	0	41	522	1547	2783	4809	4604	1214	133	1	0
	CDH80	5444	0	0	3	120	437	896	1839	1819	315	15	0	0
Wind	WSAvg	3.5	3.1	3.4	3.8	4.1	4.2	3.8	3.7	3.7	3.5	3.2	2.8	2.9
Precipitation	PrecAvg	36.4	0.3	0.4	0.8	1.8	3.3	3.8	9.3	8.6	6.0	1.6	0.7	0.2
	PrecMax	52.1	1.0	0.9	1.9	5.2	10.4	10.2	18.1	27.3	15.1	3.4	3.1	0.6
	PrecMin	22.6	0.0	0.0	0.2	0.5	1.1	1.6	2.7	2.1	1.2	0.2	0.0	0.0
	PrecStd	8.5	0.2	0.2	0.4	1.1	2.3	2.0	4.0	5.2	3.5	0.8	0.6	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.1	66.5	78.7	87.8	91.5	93.2	96.1	96.1	91.8	82.2	71.7	59.4
		MCWB	46.9	51.8	60.2	65.5	67.7	72.2	77.5	76.4	74.2	66.4	59.1	48.6
	2%	DB	55.3	62.7	73.7	83.4	87.7	89.9	93.1	93.7	87.1	77.6	68.1	56.4
		MCWB	45.4	50.5	57.9	63.8	67.0	71.8	76.8	76.4	73.2	65.2	58.2	47.2
	5%	DB	52.3	59.5	70.1	79.9	84.8	87.7	90.9	91.3	83.7	74.5	65.4	54.1
		MCWB	44.1	49.1	56.7	62.8	66.6	71.7	76.3	76.1	71.5	64.3	57.0	46.5
	10%	DB	49.9	56.5	66.6	76.4	81.8	85.3	88.6	88.7	80.5	71.7	62.9	52.1
		MCWB	43.1	47.9	54.9	62.1	65.7	71.2	75.9	75.5	70.3	63.5	56.0	45.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.6	53.5	63.8	69.8	72.7	77.3	80.7	80.2	78.4	69.8	61.5	52.1
		MCDB	55.7	63.4	76.8	80.6	81.9	84.5	90.8	90.5	87.1	76.4	67.6	56.0
	2%	WB	46.8	51.9	59.8	66.8	70.8	75.7	79.3	79.0	75.6	67.8	59.9	50.0
		MCDB	53.0	60.0	69.3	77.7	80.6	83.1	88.4	88.5	82.5	73.0	65.2	53.2
	5%	WB	45.4	50.3	57.8	64.7	69.3	74.5	78.3	78.0	73.7	66.3	58.8	48.3
		MCDB	50.6	57.5	67.2	75.4	79.0	82.4	86.8	87.0	80.3	71.9	63.1	52.4
	10%	WB	44.1	48.8	56.2	63.2	68.0	73.3	77.3	77.0	71.8	64.8	57.6	46.9
		MCDB	48.6	55.2	64.9	73.5	77.0	81.6	85.4	85.5	78.0	70.3	61.5	50.8
Mean Daily Temperature Range	5% DB	MDBR	9.7	10.4	12.5	14.1	14.1	12.3	12.0	12.4	10.1	9.4	9.5	9.5
		MCDBR	14.7	16.2	18.6	19.9	19.5	17.3	16.1	16.4	14.9	14.1	14.0	13.1
		MCWBR	7.2	7.0	6.8	6.0	5.4	4.7	4.8	4.7	4.2	4.7	5.9	6.5
	5% WB	MCDBR	12.8	13.9	16.0	16.6	14.7	13.1	13.6	13.9	11.8	11.1	10.8	10.6
MCWBR		6.9	6.7	6.5	6.0	5.6	4.6	5.1	5.0	4.5	4.6	5.5	5.9	
Clear-Sky Solar Irradiance	taub	0.656	0.708	0.732	0.715	0.674	0.642	0.665	0.694	0.701	0.714	0.699	0.655	
	taud	1.502	1.439	1.435	1.507	1.632	1.728	1.704	1.627	1.589	1.520	1.501	1.530	
	Ebn at Noon	171	179	190	203	213	220	214	205	195	176	160	162	
	Edh at Noon	77	90	96	92	82	74	76	81	81	81	76	72	
All-Sky Solar Radiation	RadAvg	583	685	947	1228	1355	1292	1377	1317	918	712	594	548	
	RadStd	67	117	115	146	160	153	219	192	118	111	68	80	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	+1.17	-3.31	N/A	N/A	-0.77	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	+0.47	N/A	-1.66	N/A	N/A	N/A	-63	-113	+189	+88

Nomenclature: See separate page